FLAMELENS

Physics Foundations for the Three Combustion Models

Equations, NASA Data Mapping, and a Small-Data Training Strategy

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A practical technical guide for

Flame Extinction | **Flame Spread** | **Soot / Smoke**

Purpose:	Build scientifically defensible, data-aware ML models from NASA combustion data.
Project inputs:	Raw NASA CSV field documentation and FlameLens ML strategy.
Status:	Research and implementation guide; not a validated combustion solver.
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Internal project document. Equations are reduced-order physical relationships for feature design, interpretation, and model checks. They must not be mistaken for a complete CFD or chemical-kinetics model.

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1 Purpose, scope, and the most important design decisions

FlameLens is intended to combine machine learning (ML), NASA experimental evidence, and physics-based interpretation. The project strategy correctly avoids inventing labels or displaying a prediction merely because a website input form exists. The present document translates that strategy into the physics and equations needed for three model tracks, with a specific emphasis on learning from limited and heterogeneous datasets.

The project field document identifies three broad categories: flame extinction, flame spread, and soot/smoke. It also distinguishes pre-test inputs from post-test observations, for example initial droplet diameter versus visible flame extinction diameter, initial oxygen versus final oxygen, sample thickness versus spread rate, and fuel rate versus burn time [1]. The strategy document recommends FLEX as the first extinction model, an experiment-explorer workflow for flame spread until a consistent measured target is found, and target identification before training a SPICE soot/smoke model [2]. This guide preserves those decisions while adding a path toward training all three models responsibly.

1.1 Three model tracks, but more than three physical regimes

The product can have three user-facing modules, but the training design should respect the physical regimes inside each module:

1. **Flame Extinction:** FLEX droplet combustion and PSI-159 CFI-G spherical gas-diffusion-flame experiments are related by the broad question of flame survival, but they are not the same physical system. The initial implementation should use separate model branches or separate experts, not blindly pool their rows into one table. FLEX studies burning liquid droplets and near-limit extinction; PSI-159 CFI-G studies spherical diffusion flames, including normal and inverse configurations and cool-flame conditions [3, 5].
2. **Flame Spread:** BASS-II and SAFFIRE provide valuable solid-fuel experiments, but sample geometry, material, airflow, oxygen, ignition procedure, and experimental scale differ. A shared model is possible only after compatible targets and units are established, with experiment/source grouping retained for validation [6, 7, 8].
3. **Soot / Smoke:** SPICE is specifically concerned with laminar smoke-point properties of non-buoyant jet diffusion flames under co-flow conditions, including burner diameter and co-flow velocity [9, 10]. A smoke-point length, luminous flame length, soot volume fraction, and soot-emission label are different targets and must not be treated as interchangeable.

Caution. The project documents do not, by themselves, establish that every CSV contains a complete, consistently measured target for every model. A field name being listed in the documentation is not proof that it exists in every experiment file, is populated, or is comparable across missions. The target audit in Section 7 is therefore mandatory before fitting each model.

1.2 A prediction must represent a defined scientific question

Before training, write each prediction task in one sentence. Examples:

- **Extinction classification:** Given conditions known before or at a defined prediction time, will the specified flame configuration extinguish within the experiment's observation window?
- **Flame-spread regression:** Given material, geometry, atmosphere, ignition, and flow conditions, what measured spread rate will occur over a precisely defined interval?
- **Smoke-point regression:** Given fuel and burner/co-flow conditions, what is the measured smoke-point length under the SPICE measurement definition?

The wording determines the correct label, allowed inputs, treatment of tests that end before an event, and evaluation metric.

2 Combustion physics shared by the three models

2.1 Conservation laws

All three problems are governed by conservation of mass, momentum, species, and energy. A full reacting-flow solver solves coupled partial differential equations. FlameLens does not need to solve the full equations for every user prediction, but the equations explain which input variables matter and help create physically meaningful features.

Mass conservation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \tag{1}$$

Species conservation For species i with mass fraction Y_i,

$$\frac{\partial (\rho Y_i)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = -\nabla \cdot \mathbf{j}_i + \dot{\omega}_i, \tag{2}$$

where $\mathbf{j}_i$ is the diffusive flux and $\dot{\omega}_i$ is the chemical production or consumption rate.

Energy conservation A useful schematic form is

$$\rho c_p \frac{DT}{Dt} = \nabla \cdot (k \nabla T) - \sum_i h_i \dot{\omega}_i - \nabla \cdot \mathbf{q}_{\text{rad}} + \Phi, \tag{3}$$

where k is thermal conductivity, h_i is species enthalpy, $\mathbf{q}_{\text{rad}}$ is radiative heat flux, and Φ represents viscous-dissipation terms where relevant. The detailed sign convention for chemical heat release depends on how species enthalpies and reaction rates are defined; the key idea is that chemical heat release competes with conduction, convection, and radiation.

2.2 Stoichiometry and mixture strength

For a hydrocarbon fuel $\mathrm{C}_x\mathrm{H}_y$ burning completely,

$$\mathrm{C}_x\mathrm{H}_y + \left(x + \frac{y}{4}\right) \mathrm{O}_2 \longrightarrow x\,\mathrm{CO}_2 + \frac{y}{2}\,\mathrm{H}_2\mathrm{O}. \tag{4}$$

The stoichiometric oxygen requirement per mole of fuel is therefore $\nu_{\mathrm{O}_2,\mathrm{st}} = x + y/4$. For oxygenated fuels, the oxygen already contained in the fuel must also be included; do not apply the hydrocarbon-only expression unchanged.

The equivalence ratio is

$$\phi = \frac{(\dot{m}_f / \dot{m}_{ox})}{(\dot{m}_f / \dot{m}_{ox})_{st}}, \tag{5}$$

where $\dot{m}_f$ and $\dot{m}_{ox}$ are fuel and oxidizer mass-flow rates. In premixed systems $\phi < 1$ is lean and $\phi > 1$ is rich. In a diffusion flame the local mixture varies in space, so ϕ based on total stream flow is a useful global descriptor, not a substitute for the local mixture field.

For a two-stream diffusion flame, a stoichiometric mixture fraction Z_{st} locates the stoichiometric surface. One common two-stream expression is

$$Z_{st} = \left(1 + \frac{Y_{F,\infty} W_O \nu_O}{Y_{O,\infty} W_F \nu_F}\right)^{-1}, \tag{6}$$

where Y denotes stream mass fraction, W molecular weight, and ν the stoichiometric coefficient under the chosen fuel/oxidizer convention. Different mixture-fraction definitions exist; the implementation must use the same definition as the NASA field or derive it consistently from documented stream compositions [11].

2.3 Adiabatic flame temperature

The adiabatic flame temperature T_{ad} is the equilibrium or prescribed-product temperature obtained from an enthalpy balance with no heat loss. In a simplified constant-pressure enthalpy formulation,

$$\sum_r n_i \left[\Delta h^\circ_{f,i} + \int_{T^\circ}^{T_r} c_{p,i}(T)\ \mathrm{d}T\right] = \sum_p n_j \left[\Delta h^\circ_{f,j} + \int_{T^\circ}^{T_{ad}} c_{p,j}(T)\ \mathrm{d}T\right]. \tag{7}$$

The product composition depends on the chemical-equilibrium or frozen-chemistry assumption. Thus, T_{ad} is not a universal lookup value: fuel, oxidizer, dilution, pressure, and the thermochemical model matter. A precomputed T_{ad} may be a useful physics feature only if it was calculated from pre-test conditions using a documented and consistent method.

2.4 Why microgravity changes the physics

On Earth, buoyancy can drive natural convection around a flame. In microgravity, buoyancy-driven flow is strongly reduced, so diffusion, imposed forced flow, transient motion, and radiation become relatively more important. NASA describes microgravity flames as rounded or nearly spherical in low-momentum conditions because the buoyant plume is suppressed [12].

A useful indicator is the Grashof number,

$$\mathrm{Gr} = \frac{g\,\beta\,\Delta T\,L^3}{\nu^2}, \tag{8}$$

where g is effective gravity, β thermal expansion coefficient, ν kinematic viscosity, and L a characteristic length. As effective gravity decreases, buoyancy becomes less important. For imposed flow, the Reynolds and Peclet numbers are

$$\mathrm{Re} = \frac{UL}{\nu}, \qquad \mathrm{Pe} = \frac{UL}{\alpha}, \tag{9}$$

where U is a characteristic velocity and α thermal diffusivity. These dimensionless groups help compare experiments; they do not make physically different experiments automatically equivalent.

A Damköhler number compares a flow/transport timescale with a chemical timescale:

$$\mathrm{Da} = \frac{\tau_{flow}}{\tau_{chem}}. \tag{10}$$

When chemistry is too slow relative to transport or strain, a flame may fail to sustain itself. There is no single universal critical Da for all three FlameLens tasks; the threshold depends on geometry, chemistry, radiation, and the definition of extinction.

3 Model 1 — Flame Extinction

3.1 What physical problem is being modelled?

Extinction is the loss of a self-sustaining flame. In the FlameLens data, two different families are relevant:

- **FLEX / PSI-69:** liquid-fuel droplet combustion in a controlled gaseous environment. The project field document lists fuel, ambient pressure, initial gas composition, initial droplet diameter, and measured outcomes such as visible flame extinction diameter and burn time [1]. NASA describes FLEX as studying near-limit droplet flames, suppressant effectiveness, and the limiting oxygen index [4].
- **PSI-159 / CFI-G:** spherical gas-phase diffusion flames, including normal and inverse configurations, cool-flame conditions, reactant dilution, flow rates, and pressure [5]. This is not a liquid-droplet dataset, even though the high-level product category is also extinction.

Caution. Do not train one ordinary tabular model by simply concatenating FLEX droplets and PSI-159 gas flames. Keep separate models/experts or use an explicitly validated regime-conditioned architecture. A source label alone does not repair a mismatch in physical geometry or target definition.

3.2 Droplet combustion: evaporation and the d^2 law

A burning droplet receives heat from the flame and surrounding gas. Fuel evaporates from its surface, mixes with oxygen, and reacts in a surrounding flame. Under quasi-steady, single-component, spherical assumptions, the classical d^2 law approximates the reduction in squared droplet diameter:

$$d^2(t) = d_0^2 - Kt, \tag{11}$$

where d_0 is the initial diameter and K is the burning-rate constant. The idealized lifetime to zero diameter is

$$t_{life} \approx \frac{d_0^2}{K}. \tag{12}$$

This is a physical scaling relationship, not a guaranteed formula for the observed burn time in a transient, multi-component, radiating, or near-extinction experiment [13].

A simplified mass-transfer expression for a spherical droplet is

$$\dot{m}_v \approx \pi d\, \rho_g D_v\, \mathrm{Sh}\, \ln(1 + B_M), \tag{13}$$

where ρ_g is gas density, D_v vapour diffusivity, Sh the Sherwood number, and B_M the Spalding mass-transfer number. For a simple single-fuel representation,

$$B_M = \frac{Y_{F,s} - Y_{F,\infty}}{1 - Y_{F,s}}, \tag{14}$$

with $Y_{F,s}$ the fuel-vapour mass fraction at the surface and $Y_{F,\infty}$ the far-field value. Under the same quasi-steady assumptions,

$$K \approx \frac{4\rho_g D_v \mathrm{Sh}}{\rho_l} \ln(1 + B_M), \tag{15}$$

where ρ_l is liquid density. For a purely diffusive stagnant sphere, Sh is often approximated by 2; forced convection changes the transfer number.

A simplified droplet energy balance is

$$m_d c_{p,l} \frac{\mathrm{d}T_d}{\mathrm{d}t} = hA_d(T_g - T_s) + \dot{Q}_{rad,abs} - \dot{m}_v L_v, \tag{16}$$

where h is a convective heat-transfer coefficient, A_d droplet surface area, T_g gas temperature, T_s surface temperature, $\dot{Q}_{rad,abs}$ absorbed radiative heat, and L_v latent heat of vaporization. This balance explains why droplet size, fuel properties, ambient composition, pressure, and diluent properties can matter.

Useful physics-derived features

- d_0^2 rather than only d_0, because idealized droplet lifetime scales with the square of the initial diameter.
- A mass-transfer or heat-transfer indicator if the required fuel properties and gas-state variables are actually available.
- Initial oxygen mole fraction and diluent fractions, retained separately rather than collapsed into a guessed scalar.
- Pressure, because gas density, diffusivity, reaction rates, and transport change with pressure.

3.3 Extinction mechanisms

A flame can disappear for more than one physical reason. The model should not assume that every extinction event is governed by one universal threshold.

1. Oxygen-limited or dilution-driven extinction Reducing oxygen availability or adding inert diluent can reduce reaction rates and flame temperature. Nitrogen and helium are not interchangeable diluents: they have different thermal and transport properties. Carbon dioxide may also participate in radiative and chemical effects. The FLEX research programme explicitly considers oxygen mole fraction and suppressant/diluent effects [4].

2. Kinetic or transport extinction A flame may extinguish when chemical heat release cannot keep pace with transport, strain, or dilution. The Damköhler number in Eq. (10) is a useful conceptual feature. If a characteristic chemical time can be estimated from a documented kinetic model, it can be included; do not invent a chemistry time from a dataset identifier.

3. Radiative extinction A useful gray-body scaling for radiative heat transfer is

$$\dot{Q}_{rad} \approx \varepsilon \sigma A \left(T_f^4 - T_{sur}^4 \right), \tag{17}$$

where ε is an effective emissivity, σ the Stefan–Boltzmann constant, A an effective radiating area, T_f flame temperature, and T_{sur} surroundings temperature. Real gas flames are participating media; Eq. (17) is a reduced-order approximation, not a full spectral radiation model.

For spherical gaseous diffusion flames in microgravity, radiative extinction has been observed and modelled; NASA's report notes that radiative loss and temperature evolution can be central to transient extinction [14]. Therefore, oxygen alone should not be treated as a complete predictor.

4. Cool-flame and visible-flame distinction FLEX-related research has reported conditions in which chemical activity can continue after the visible flame disappears [12]. Therefore, the label "visible flame extinguished" must not automatically be interpreted as "all combustion chemistry stopped." If the dataset distinguishes hot ignition, cool flames, or other regimes, preserve those distinctions and verify when each field becomes known.

3.4 PSI-159: normal and inverse spherical diffusion flames

For the CFI-G branch, the flame is a spherical diffusion flame. In a **normal** configuration, fuel is supplied into an oxidizing environment; in an **inverse** configuration, oxidizer is supplied into a fuel-containing environment. The location of the stoichiometric surface and the transport of heat and species differ between these configurations.

The two-stream Z_{st} in Eq. (6) is useful for describing the stoichiometric surface, provided the project calculation matches the NASA definition and uses the correct stream compositions. The

adiabatic flame temperature from Eq. (7) helps describe the thermochemical potential of the mixture. Both may be useful inputs or derived features when defined before the predicted event.

Caution. The raw-field document lists a field named `TFP` but explicitly says its exact physical meaning must be verified before it is used as a numerical ML feature [1]. Keep it out of the model until the original NASA data dictionary or paper defines it. Do not guess what the abbreviation means.

3.5 Target definitions for the extinction model

Possible targets, in order of practicality, are:

1. **Binary event target:** extinction versus sustained/non-extinction, but only if the experiment record explicitly supports both labels and the observation window is comparable.
2. **Time-to-extinction:** useful when event times are recorded. A test that ends before extinction is a right-censored observation, not an ordinary negative example.
3. **Extinction diameter or burn-rate regression:** only when a consistent measured value exists and it is the intended prediction target.

For time-to-event data, survival analysis is often more defensible than forcing every run into a binary class. Let T be extinction time and $\delta = 1$ indicate that extinction was observed before the test ended. A survival model estimates

$$S(t \mid \mathbf{x}) = P(T > t \mid \mathbf{x}), \tag{18}$$

while retaining censored runs. A Cox model or a regularized accelerated-failure-time model can be a baseline if the number of independent experiments is adequate.

3.6 Recommended inputs and leakage controls

3.7 Small-data modelling plan for extinction

1. Establish a FLEX-only baseline with a regularized logistic regression for binary classification. Compare with a shallow Random Forest. Do not begin with a neural network.
2. If there are few independent experimental runs, prefer grouped cross-validation by experiment/test/sample. If several droplets belong to one run, keep the entire run in one fold.
3. If the target is time-to-extinction with censoring, test a simple survival model rather than discarding censored runs.
4. Build a separate CFI-G model only after the PSI-159 labels and pre-test feature definitions are confirmed. Include normal/inverse flame configuration as a physical regime, not as an arbitrary text token.
5. Report probability calibration and uncertainty. A nominal probability of 0.8 should be interpreted as an 80% event frequency only if calibration has been checked on independent groups.

4 Model 2 — Flame Spread

4.1 Physical picture

A solid fuel does not usually burn by simply igniting at every point. Heat from the existing flame preheats the unburned solid ahead of the flame front. The solid decomposes or melts/pyrolyses, releasing combustible gases. Those gases mix with oxygen, react, and feed heat back to the solid.

Field family	Potential role	Important condition
Fuel / fuel composition	Input	Encode fuel identity categorically; add molecular descriptors only from verified sources.
Initial oxygen and initial diluent composition	Input	Preserve mole fraction versus volume percent distinctions; convert only with justified assumptions.
Ambient pressure	Input	Convert units consistently (e.g. mmHg, bar, Pa).
Initial droplet diameter	Input for FLEX	Do not substitute visible extinction diameter.
Fuel and oxygen flow rates	Input for gas-flame branch	Check that flow rates refer to the actual burner streams and compatible units.
T_{ad}, Z_{st}	Derived or recorded input	Use only if calculated from pre-test composition and the same documented convention.
Visible flame extinction diameter, burn time, final gas composition	Target / outcome / post-test observation	Never use as pre-test predictors for the outcome they describe.
Test identifiers, timestamps, PI, comments	Provenance / grouping	Retain for traceability and grouped validation; ordinarily exclude from predictive features.

Table 1: Extinction inputs and leakage safeguards based on the field documentation [1].

Flame spread continues when forward heat feedback is sufficient to prepare new fuel faster than heat is lost.

In microgravity, buoyant flow is suppressed. The imposed airflow, if present, can dominate oxygen delivery and heat/species transport. BASS-II studied solid samples in low-velocity forced flow, varying airflow, oxygen concentration, fuel thickness and flow configuration [6]. SAFFIRE observations show that microgravity flame behaviour and scale can differ from familiar upward-spreading 1-g flames; concurrent flow can reach a limiting flame length and steady spread behaviour under some conditions [8].

4.2 Opposed flow and concurrent flow

- **Opposed flow:** the imposed oxidizer flow travels against the direction in which the flame front spreads over the solid. The flame is stabilized near the leading edge and can be sensitive to gas-phase transport and blow-off at high flow.
- **Concurrent flow:** imposed flow and flame spread move in the same direction. The flame can interact with a developing boundary layer and, in some microgravity configurations, approach a steady flame size and spread rate.

The field document contains both `Flow Direction` and `Direction`; normalize them to a common category only after checking the original dataset definitions [1]. Do not assume that a text value such as “forward” means the same reference direction in every mission.

4.3 Solid-phase energy balance

A useful one-dimensional reduced-order surface-energy balance is

$$\rho_s c_s \delta_{eff} \frac{\partial T_s}{\partial t} = k_s \delta_{eff} \frac{\partial^2 T_s}{\partial x^2} + q''_{flame\to s} + q''_{ext} - q''_{rad} - q''_{conv} - \dot{m}''_p L_p. \tag{19}$$

Here ρ_s, c_s, k_s are solid density, heat capacity, and conductivity; δ_{eff} is an effective thermal thickness; $q''_{flame\to s}$ is flame-to-solid heat feedback; q''_{ext} is external heating; q''_{rad} and q''_{conv} are radiative and convective losses; $\dot{m}''_p$ is the mass-loss rate from pyrolysis per unit area; and L_p is an effective energy requirement associated with decomposition/vaporization. Equation (19) is a modelling template. For thermally thick samples, a resolved solid conduction model may be required instead of one effective thickness.

Radiative loss can be approximated by

$$q''_{rad} = \varepsilon_s \sigma \left(T_s^4 - T_{sur}^4 \right), \tag{20}$$

with an effective emissivity ε_s. The actual radiation field can include flame radiation, surface reradiation, and view-factor effects.

4.4 Pyrolysis kinetics

A simple global Arrhenius form for a solid conversion variable α is

$$\frac{\mathrm{d}\alpha}{\mathrm{d}t} = A_p \exp\left(-\frac{E_p}{R_u T_s} \right) f(\alpha), \qquad \dot{m}''_p = \rho_s \delta_{eff} \frac{\mathrm{d}\alpha}{\mathrm{d}t}, \tag{21}$$

where A_p is a pre-exponential factor, E_p activation energy, R_u the universal gas constant, and $f(\alpha)$ a conversion function. The parameters are material-specific. A single generic pyrolysis coefficient should not be assigned to PMMA, fabric, paper, and other materials.

4.5 Flame spread rate and thermal length scales

If the flame-front position is $x_f(t)$, the instantaneous spread rate is

$$S_f = \frac{\mathrm{d}x_f}{\mathrm{d}t}. \tag{22}$$

In practice, estimate S_f from a defined interval of video-derived front positions, with a consistent image calibration and flame-front definition. If the front accelerates or becomes steady, do not report a single average without also recording the interval.

A basic thermal-diffusion timescale over a preheat length ℓ_p is

$$\tau_{th} \sim \frac{\ell_p^2}{\alpha_s}. \tag{23}$$

Equating the order of a front-advection time ℓ_p / S_f with τ_{th} gives the scaling

$$S_f \sim \frac{\alpha_s}{\ell_p}. \tag{24}$$

This is a scaling estimate, not a universal predictive law. The actual preheat length depends on gas-phase flame structure, airflow, geometry, heat feedback, radiation, and chemistry.

Useful dimensionless features include

$$\mathrm{Pe}_g = \frac{UL}{\alpha_g}, \qquad \mathrm{Re}_g = \frac{UL}{\nu_g}, \qquad \mathrm{Fo}_s = \frac{\alpha_s t}{L_s^2}, \qquad \mathcal{R}_{rad} = \frac{q''_{rad}}{q''_{flame\to s} + \epsilon_q}. \tag{25}$$

Here L and L_s are explicitly chosen characteristic lengths, and ϵ_q is a small numerical stabilizer only if the denominator can approach zero. Record the exact definitions used in code so the feature is reproducible.

4.6 What controls spread?

Factor	Physical effect	Candidate field family
Material chemistry	Determines decomposition products, ignition temperature, heat of pyrolysis, and char/ash behaviour.	Material; fuel sample material.
Oxygen concentration	Changes oxidation rate, flame temperature, and ability to sustain the gas-phase flame.	Percent O_2; calibrated initial O_2.
Airflow speed	Controls oxygen transport, flame residence time, and convective heat transfer.	Air Flow; Air Rate; Air display, after units/meaning are verified.
Flow direction	Changes flame stabilization and coupling between flame and unburned fuel.	Flow Direction; Direction.
Thickness and geometry	Change thermal inertia, conduction paths, available fuel, and radiative losses.	Sample thickness, length, width.
Ignition conditions	Affect the initial hot zone and whether a self-sustaining flame is established.	Ignition power and ignition time.
Gravity / scale	Changes buoyancy and can alter flow structure and flame dimensions.	Experiment-level gravity metadata; source/geometry metadata.

Table 2: Physics-to-feature map for Flame Spread, based on the project field document and NASA experiments [1, 6, 7, 8].

4.7 Targets and sources

The initial training target should be a measured spread rate with a documented definition, unit, image-processing procedure, and time interval. The project strategy notes that the currently assembled BASS-II and SAFFIRE tables do not yet provide a clean, consistent spread-rate target and recommends investigating additional BASS/BASS-II analysed data and PSI-62 work before claiming a general predictor [2]. That is a data-readiness issue, not a reason to invent a target.

Possible target hierarchy:

1. Flame-front spread rate S_f over a specified interval (preferred regression target if measured consistently).
2. Binary outcome: sustained spread versus extinction, only if the observation protocol supports a consistent label.
3. Flame length or pyrolysis length, but only as a separate target with its own measurement definition.

4.8 Small-data modelling plan for flame spread

1. Start with a physics-informed baseline: a regularized regression or generalized additive model using material, oxygen, airflow, thickness, and flow direction. Add only verified material properties.

2. Add a Gaussian Process (GP) or shallow tree ensemble if the number of independent tests supports it. GPs can provide useful uncertainty estimates on small smooth datasets, but the kernel and noise model must be checked.
3. Use material- and experiment-aware validation. A random row split can place repeated observations from the same test in training and testing and exaggerate performance.
4. Keep BASS-II and each SAFFIRE mission identifiable for domain-shift analysis. Compare a pooled model against source-specific models; pooling is justified only if it improves leave-one-source-out performance.
5. If labels remain sparse, deploy the module as an experiment explorer with nearest NASA tests and measured outcomes. This is still useful functionality and must not be labelled as an ML spread-rate prediction.

5 Model 3 — Soot / Smoke

5.1 What SPICE measures

SPICE studies the smoke point of non-buoyant jet diffusion flames in co-flowing air. NASA identifies burner diameter and co-flow velocity as variables affecting luminous flame length and smoke point [9]. The SPICE project summary also reports tests with several gaseous fuels and more than 70 identified smoke points in the historical investigation [10].

The project's field documentation lists fuel, nozzle diameter, fuel rate, hydrocarbon fuel rate, and air rate as available SPICE field families, while oxygen concentration, pressure, and temperature are not directly represented in the listed SPICE input fields [1]. Do not silently fabricate those missing variables.

5.2 Fuel-air mixing and global stoichiometry

For a gaseous fuel supplied at standard volumetric flow $\dot{V}_{std}$, convert to molar flow using the stated standard conditions:

$$\dot{n}_f = \frac{p_{std}\dot{V}_{std}}{R_u T_{std}}, \qquad \dot{m}_f = \dot{n}_f W_f. \tag{26}$$

This conversion is valid only when $\dot{V}_{std}$ is truly reported at the same standard pressure and temperature. Different instruments and datasets sometimes use different reference conditions. Preserve the raw value and store the conversion assumptions.

For a circular nozzle of diameter D_n,

$$A_n = \frac{\pi D_n^2}{4}, \qquad U_f = \frac{\dot{V}_f}{A_n}. \tag{27}$$

Nozzle exit velocity can be a useful derived feature if volumetric flow is measured at the nozzle conditions. Do not combine a standard volumetric flow with the nozzle area as though it were the actual hot-gas volume flow without correcting pressure and temperature.

The global equivalence ratio can be estimated from measured fuel and oxidizer streams using Eq. (5), provided their flow rates are compatible and the fuel composition is known. In diffusion flames, local mixture fraction and residence time govern where soot chemistry occurs; global equivalence ratio alone is not sufficient.

5.3 Soot formation and oxidation

Soot formation is not a single reaction. A useful conceptual sequence is:

1. Fuel pyrolysis and formation of unsaturated/aromatic precursors.
2. Particle inception or nucleation.
3. Surface growth as gas-phase species add carbon to particles.
4. Coagulation and agglomeration, which change particle number and size distribution.
5. Oxidation, which removes soot carbon.

These stages are widely used in soot-model descriptions [15].

A reduced soot mass-fraction transport equation is

$$\frac{\partial(\rho Y_s)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_s) = \nabla \cdot (\rho D_s \nabla Y_s) + \dot{\omega}_s, \tag{28}$$

with a schematic net source

$$\dot{\omega}_s = \dot{\omega}_{inception} + \dot{\omega}_{growth} - \dot{\omega}_{oxidation}. \tag{29}$$

Coagulation is important for particle number and size distribution, but it is approximately mass-conserving by itself; it should not be inserted as an independent net soot-mass source without a carefully defined moment model.

The balance in Eq. (29) explains why both fuel-rich pyrolysis conditions and oxidizing conditions matter. More fuel does not necessarily mean soot output rises linearly: soot can grow in a rich region and later be oxidized before leaving the flame.

5.4 Smoke point, flame length, and residence time

A smoke point is an experimentally defined condition at which soot becomes visible as emitted smoke from the flame. It is not the same as luminous flame length, soot volume fraction, total soot mass, or a binary soot-presence label. Confirm the exact NASA/SPICE measurement and post-processing definition before choosing the target.

A simple residence-time feature is

$$\tau_{res} \sim \frac{L_f}{U_c}, \tag{30}$$

where L_f is a characteristic flame length and U_c a representative convective velocity. This is only a reduced estimate because the velocity varies spatially and the flow can be entraining and developing.

Experiments on non-buoyant co-flowing microgravity diffusion flames reported an empirical smoke-point-length scaling of the form

$$L_{sp} \propto D_n^{-0.91} U_{air}^{0.41}, \tag{31}$$

for the investigated burners, fuels, and co-flow conditions [16]. This is a useful prior for feature design and qualitative checks, not a universal law to impose on every SPICE record. The fitted exponents are study- and range-dependent.

5.5 Candidate targets and input fields

- **Direct candidate inputs from the project field document:** fuel, nozzle diameter, fuel rate, hydrocarbon fuel rate, and air rate [1].

Candidate	Meaning	Readiness rule
Smoke-point length	Distance/height at which smoke first appears, using the NASA definition.	Preferred if a consistent analysed measurement is available.
Luminous flame length	Visible flame dimension.	A separate regression task; not a substitute for smoke-point length.
Soot volume fraction	Local/field soot loading derived from diagnostics.	Requires analysed optical/diagnostic labels and calibration metadata.
Soot-emission classification	Whether smoke is emitted under a defined observation rule.	Requires explicit labels and consistent visual/measurement criteria.

Table 3: Candidate soot/smoke targets. Select exactly one primary target for the first supervised model.

- **Useful derived features:** mass-flow rate after standard-condition conversion, nozzle exit velocity where defensible, fuel-to-air ratio, global equivalence ratio, D_n-based geometric features, and residence-time proxies if velocity/length are actually measured.
- **Unavailable in the listed SPICE input fields:** oxygen concentration, pressure, and temperature. Add them only if the matching NASA source data provides them for the exact run [1].

5.6 Small-data modelling plan for soot/smoke

1. Locate the analysed SPICE smoke-point or soot diagnostics before model fitting. The project strategy correctly calls this the first step [2]. NASA's PSI repository also describes a SPICE data-analysis investigation that used image processing and soot diagnostics; examine whether its analysed outputs can supply reliable labels [17].
2. Train a simple baseline first: ridge regression for a continuous smoke-point target, or regularized logistic regression for a binary target. Compare with a shallow Random Forest or GP only after checking data coverage.
3. Split by experimental run, fuel/burner configuration, or campaign. If many image frames come from one flame, frames from a single test must not appear in both training and test partitions.
4. Keep fuel identity, fuel composition, nozzle geometry, and co-flow setting explicit. Do not let a model use test IDs, camera names, or filenames as shortcuts.
5. If soot volume fraction is reconstructed from images, retain calibration, camera, exposure, and image-processing provenance. Image-derived labels can carry systematic errors and are not independent samples merely because they contain many pixels.

6 Building useful models when the dataset is small

Small experimental datasets are common in spacecraft combustion because each test is expensive, highly controlled, and sometimes mission-limited. The solution is not to manufacture rows. It is to extract more valid information from each experiment, encode known physics, and quantify uncertainty honestly.

6.1 First principle: increase information, not row count

A dataset with 500 correlated video frames from one experimental run may contain far less independent information than 50 genuinely separate tests. Count independent tests, campaigns, material/fuel configurations, and measurement conditions. Record both the number of rows and the number of independent experimental groups.

6.2 Physics-informed feature engineering

Use a small number of features with a defensible physical meaning. Candidate examples include:

- Extinction: d_0^2, oxygen/diluent fractions, pressure, flow rates, Z_{st}, and a documented T_{ad}.
- Spread: oxygen level, airflow, material, thickness, thermal diffusivity/conductivity where verified, flow direction, and dimensionless groups such as Pe and Fo.
- Soot: fuel mass flow, nozzle diameter, co-flow velocity, fuel-to-oxidizer ratio, and validated residence-time or stoichiometric descriptors.

Avoid adding dozens of derived variables that all encode the same information. With small N, a very wide feature set increases variance and makes explanations unstable.

6.3 Recommended model ladder

Stage	Method	Why / when
1. Baseline	Mean/median predictor or majority-class predictor	Establishes the minimum benchmark; a complex model must beat it.
2. Interpretable	Ridge/Lasso/Elastic Net; logistic regression	Good for low sample counts, stable baselines, and checking whether features contain signal.
3. Nonlinear	Shallow Random Forest / Extra Trees / gradient boosting with conservative depth	Captures interactions, but tune only a few hyperparameters and validate by group.
4. Small-data probabilistic	Gaussian Process or Bayesian regression/classification	Can model smooth relationships and uncertainty when feature count and sample count are manageable.
5. Physics hybrid	Physics baseline plus a learned residual	Uses ML for systematic corrections rather than asking it to rediscover known physical trends.

Table 4: A model ladder. Not every task should progress to every stage.

A physics-hybrid regression can be written as

$$y_{obs} = y_{phys}(\mathbf{x}; \boldsymbol{\theta}) + f_{ML}(\mathbf{x}) + \epsilon, \tag{32}$$

where y_{phys} is a reduced-order physical prediction, f_{ML} learns the residual discrepancy, and ϵ represents measurement noise and unmodelled effects. Use this only when the physics baseline is meaningful and its assumptions are documented. A poor physics model can create a misleading residual target.

6.4 Cross-validation that respects experiments

Do not use random row splitting when rows are repeated measurements, frames, droplets in one test, or sequential observations from one sample. Use:

- **GroupKFold:** keep all records from the same test/experiment/sample in one fold.

- **Leave-One-Experiment-Out:** useful when the number of independent experiments is small and each experiment is a meaningful group.
- **Leave-One-Source-Out:** tests whether a model trained on some missions generalizes to a different source, but only when input and target definitions are compatible.
- **Nested group cross-validation:** use an inner grouped loop for hyperparameter selection and an outer grouped loop for performance estimation when the sample size permits it.

If there are too few independent groups to form stable folds, say so. Report per-fold scores and uncertainty rather than presenting one optimistic number as a definitive performance estimate.

6.5 Metrics

Classification Report the confusion matrix, precision, recall/sensitivity, specificity, F1-score, balanced accuracy, and ROC-AUC. For rare extinction or smoke-emission classes, also report precision–recall AUC. If the model reports probabilities, check Brier score and calibration plots. Accuracy alone can be misleading in an imbalanced dataset.

Regression Report MAE and RMSE in the physical target's units. Include R^2 only with context, because it can be unstable or negative on small held-out sets. Show predicted-versus-measured plots and residuals by fuel/material/source.

Uncertainty Distinguish three ideas:

1. **Measurement uncertainty:** uncertainty in the observed NASA measurement or image-derived label.
2. **Model uncertainty:** uncertainty due to limited data or parameter estimation.
3. **Applicability-domain / coverage:** whether the requested inputs resemble the conditions represented in training.

A model can be uncertain even inside a densely sampled region, and can be confidently wrong outside its training domain. Do not collapse all three into one unvalidated "confidence percentage."

6.6 Data coverage and nearest-experiment search

For each feature j, standardize using statistics calculated on the training fold only:

$$z_{ij} = \frac{x_{ij} - \mu_{j,train}}{s_{j,train}}. \tag{33}$$

A simple distance to a training record i is

$$d(\mathbf{x}, \mathbf{x}_i) = \sqrt{\sum_{j=1}^{p} w_j (z_j - z_{ij})^2}, \tag{34}$$

where weights w_j reflect feature relevance and scaling. For mixed categorical/numerical data, use a mixed-type distance or separate filtering by regime/material/fuel before numerical distance calculation. Do not treat categorical fuel names as ordinal numbers.

Define coverage using multiple checks: distance to neighbours, number of independent nearby experiments, feature-range violations, and whether the point belongs to a regime represented in training. Thresholds should be estimated from training/validation data, not selected to make the dashboard look favourable. A neighbour count should count independent experiments, not duplicate frames.

6.7 Data augmentation: what is safe and what is not

- **Good:** use validated physical simulations to generate structured parameter sweeps; use them for pretraining, feature design, or a physics baseline; then calibrate against real NASA experiments.
- **Potentially useful with care:** small measurement-noise perturbations based on known instrument uncertainty, provided the perturbations preserve physical constraints and are used only inside training folds.
- **Usually unsafe:** duplicating rows, random interpolation between unrelated experiments, adding arbitrary Gaussian noise, or applying generic SMOTE to combustion variables without checking stoichiometry, units, and physical feasibility.
- **Never:** label synthetic samples as NASA observations or use synthetic data in the held-out NASA test set.

A robust hybrid plan is to train on a validated simulator, then learn a small correction from the experimental data. The simulation-to-experiment gap must be measured; simulated points are not independent experimental validation.

6.8 Transfer learning and pooling across missions

Pooling across datasets can improve sample count only when the physics, feature definitions, units, and target labels are compatible. Recommended safeguards:

1. Keep an immutable source-dataset identifier and experiment identifier.
2. Normalize units while preserving the original field and unit.
3. Add source/mission as a grouping variable for validation; do not automatically make it a predictive shortcut.
4. Compare a pooled model, source-specific models, and a hierarchical model with source-level effects.
5. Use leave-one-source-out evaluation to test transfer. If that fails, restrict the model's stated domain or collect more bridge experiments.

For flame extinction, this means FLEX droplet records and PSI-159 spherical gas-flame records should remain separate experts unless a genuinely comparable target and representation are demonstrated. For soot, do not mix SPICE gas-phase soot with polymer-smoke aerosol measurements from a different investigation as though they were identical phenomena. For flame spread, align sample type, direction convention, measurement interval, and spread-rate definition before pooling.

7 Target and leakage audit

Run this audit separately for each dataset and each model branch before feature engineering.

7.1 Target audit checklist

1. Identify the exact NASA source file, version, table, and experiment identifier.
2. Locate the target column or derive the target from documented time-series/video measurements.
3. Confirm the physical meaning, units, measurement method, and observation interval.

4. Count valid and missing target values, distinct experiments, and independent groups.
5. Inspect target distribution, class imbalance, outliers, censoring, and repeated measurements.
6. Check whether records from different missions use the same definition and calibration.
7. Write down when each proposed input becomes known relative to the prediction time.
8. Freeze the final target definition and version it before model training.

7.2 Fields that should normally be excluded as pre-test predictors

The project documentation marks the following as outcomes or post-test observations in the relevant context [1]:

- Visible flame extinction diameter;
- Burning rate and burn time when predicting the pre-test outcome;
- Final oxygen, final carbon dioxide, final carbon monoxide, and "after test" composition fields;
- Flame-spread rate, spread length, and burn length when these are the target or measured consequences;
- Test-end or flame-off timestamps if they reveal the outcome rather than a pre-test condition.

Metadata such as PI, date, GMT, camera, file path, and test ID should generally be retained for traceability and grouping, not fed directly to the predictive model. There may be legitimate time-series tasks where a timestamp or a sensor reading is known at prediction time; define that use case explicitly.

7.3 Missing values and schema design

Preserve raw NASA column names and source units. Create a separate normalized table with project-level variables, unit metadata, transformation provenance, and a missingness reason. A field absent from a source is not the same as a field measured as zero. Store unavailable variables as null, as the project field document recommends [1].

Do not impute an entire absent physical dimension from a different mission unless a validated conversion exists. If missingness itself carries source information, be careful: the model may learn which experiment produced a row rather than combustion physics.

8 Practical build plan for the three models

The goal is to work toward three model modules while making model availability depend on evidence. A module may initially contain a baseline model, an experiment-search fallback, or a "not yet supported" result. That is more scientifically sound than publishing three unsupported predictions.

8.1 Phase 0 — Dataset and label audit

- Freeze the list of NASA source files and versions.
- Create a data dictionary: raw name, normalized name, unit, data type, source dataset, physical meaning, input/target/metadata status, and missingness.
- Verify PSI-69 FLEX event definitions and distinguish visible extinction from chemical extinction or cool-flame activity.

- Verify PSI-159 CFI-G normal/inverse configuration labels, target definitions, and the meaning of `TFP` before using it.
- For BASS-II/SAFFIRE, identify a consistent spread-rate target or reconstruct it from calibrated front-position measurements.
- For SPICE, retrieve and verify smoke-point/soot diagnostic labels; investigate PSI-60 analysed products where relevant.

8.2 Phase 1 — Build simple, reproducible baselines

Module	First model	Second comparison	Required fallback
Extinction / FLEX	Regularized logistic regression, or survival model if censored	Shallow Random Forest	Similar FLEX experiments and explicit low-coverage warning.
Extinction / CFI-G	Separate regularized model after label audit	Shallow nonlinear model if enough independent tests	Similar CFI-G experiments, separated by normal/inverse regime.
Flame Spread	Ridge/GAM baseline on verified spread-rate target	Gaussian Process or shallow forest	NASA experiment explorer until target is consistent.
Soot / Smoke	Ridge regression or logistic regression, depending on target	Gaussian Process or shallow forest	Similar SPICE experiments and target-readiness warning.

Table 5: Recommended initial model choices. These are starting points, not claims that the current data is already sufficient.

8.3 Phase 2 — Add physics-aware features

For every derived feature, record its formula, source fields, unit, assumptions, and whether it is available at prediction time. Unit tests should verify conversions and physical constraints. Examples include:

- Convert pressure to Pa internally; retain source pressure and source unit.
- Convert oxygen percentages to mole fractions consistently, not by mixing mass and volume fractions.
- Convert standard volumetric flow to molar/mass flow using the exact stated standard conditions.
- Calculate Z_{st} and T_{ad} only where stream compositions and thermochemistry are adequately specified.
- Estimate spread rate from calibrated flame-front position and a declared interval.

8.4 Phase 3 — Validate, calibrate, and define the domain

Use grouped evaluation, calibration checks, and an applicability-domain threshold. Set model-specific supported ranges for fuel, material, oxygen, pressure, flow, geometry, and gravity regime. Do not issue an ordinary prediction when a request is outside the verified domain; instead return similar NASA experiments and explain why the model is not being applied.

8.5 Phase 4 — Physics explanation and what-if analysis

Use SHAP or another feature-attribution method to describe what the fitted model used. Keep this separate from physical causation. The explanation layer should distinguish:

- **Model attribution:** which input features moved the prediction;
- **NASA evidence:** which measured experiments are most similar and what they observed;
- **Physics interpretation:** which mechanisms could explain the trend, with assumptions and references;
- **Uncertainty:** model uncertainty, measurement uncertainty, and data coverage.

For what-if analysis, change one variable at a time while holding the rest fixed, enforce physical bounds, and suppress results when the altered point leaves the training domain. A model response curve is not experimental proof of causality.

8.6 Phase 5 — Improve data efficiently

When the model identifies sparse regions, propose additional experiments that are both scientifically informative and feasible. Candidate selection can combine:

$$a(\mathbf{x}) = \lambda_u U(\mathbf{x}) + \lambda_g G(\mathbf{x}) + \lambda_p P(\mathbf{x}), \tag{35}$$

where U is predictive uncertainty, G is a data-gap score, P is a physics-based priority or sensitivity score, and λ values are documented weights. This is an experiment-selection heuristic, not a guarantee of optimality. Check safety, facility capability, and research priorities before treating a suggested point as an executable experiment.

9 Recommended feature and target map

Module / branch	Source family	Candidate pre-test inputs	Target / readiness note
Extinction: droplet	PSI-69 FLEX	Fuel; initial O_2; initial N_2/CO/He composition; ambient pressure; initial droplet diameter; known initial thermal/flow conditions.	Extinction event, time-to-extinction, or measured burning outcome only after exact event definitions and censoring are resolved.
Extinction: gas flame	PSI-159 CFI-G	Fuel; initial reactant composition; pressure; fuel/oxidizer flow; normal/inverse configuration; verified Z_{st} and T_{ad}.	A CFI-G-specific extinction/cool-flame outcome. Do not merge with FLEX target without a regime-aware design.
Flame Spread	PSI-25 BASS-II; PSI-98/99/100 SAFFIRE	Material; initial O_2; airflow; flow direction; thickness/length/width; ignition power/time; verified gravity metadata.	Measured spread rate or another explicitly defined outcome. Current project strategy says the assembled target needs further work.

Module / branch	Source family	Candidate pre-test inputs	Target / readiness note
Soot / Smoke	PSI-107 SPICE	Fuel; nozzle diameter; fuel rate; hydrocarbon fuel rate; air/co-flow rate; carefully derived flow/stoichiometric features.	Smoke-point length, soot volume fraction, or smoke-emission label. Choose one after analysed target verification.

Table 6: Proposed model input/target map. Based on the raw field documentation and the strategy document [1, 2], supplemented by NASA source descriptions [5, 6, 7, 9, 10].

10 Symbol glossary

Symbol	Meaning	Typical unit
d, d_0	Instantaneous and initial droplet diameter	m
K	Droplet d^2-law burning-rate constant	m^2/s
$\dot{m}_v$	Droplet fuel-vapour mass-loss rate	kg/s
B_M	Spalding mass-transfer number	dimensionless
Sh	Sherwood number	dimensionless
ϕ	Equivalence ratio	dimensionless
Z_{st}	Stoichiometric mixture fraction	dimensionless
T_{ad}	Adiabatic flame temperature	K
Da	Damköhler number	dimensionless
Gr	Grashof number	dimensionless
Re	Reynolds number	dimensionless
Pe	Peclet number	dimensionless
Fo	Fourier number	dimensionless
S_f	Flame spread rate	m/s or mm/s
α_s, α_g	Solid and gas thermal diffusivity	m^2/s
$\dot{m}''_p$	Solid pyrolysis mass-loss rate per area	$kg/(m^2s)$
L_p	Effective pyrolysis energy term	J/kg
D_n	Burner/nozzle diameter	m
U_{air}	Co-flow air velocity	m/s
L_{sp}	Smoke-point length	m
Y_s	Soot mass fraction	kg soot/kg mixture
$\dot{\omega}_s$	Net soot mass source	$kg/(m^3s)$

Symbol	Meaning	Typical unit
τ_{res}	Characteristic residence time	s

11 Final implementation rules

1. Maintain one immutable raw-data layer and one versioned normalized-data layer.
2. Maintain separate training tables for FLEX droplet extinction and PSI-159 gas-flame extinction.
3. Do not train Flame Spread until the spread-rate or other target is explicitly measured and harmonized.
4. Do not train Soot/Smoke until the SPICE smoke-point/soot target and its measurement method are verified.
5. Preserve units, source IDs, experiment IDs, and raw names; use experiment IDs for grouping, not as convenient predictive shortcuts.
6. Remove post-event fields from pre-event prediction inputs.
7. Use grouped validation and report the number of independent experiments, not only the number of CSV rows.
8. Compare against simple baselines. A complicated model is not automatically a better scientific model.
9. Report uncertainty and data coverage separately.
10. When evidence is insufficient, return NASA evidence and explain the limitation rather than fabricating a confident prediction.

Engineering takeaway. The strongest FlameLens architecture is not "three algorithms trained at any cost." It is three physics-grounded model tracks, each with a verified target, an honest small-data validation protocol, an applicability-domain check, and a NASA-evidence fallback. Build the predictive capability progressively; do not allow the web interface to imply scientific support that the data has not demonstrated.

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